

5 Controller Learning, Hybrid Regularization, and MORL Seed Stability

Block 1 established a deliberately asymmetric surrogate result: the compact v3 surrogate is the more useful rollout environment for reinforcement learning (RL), whereas the calibrated v3.5 RC-NeuralODE is the more accurate predictive digital twin. Block 2 tests the controller-side consequence of that asymmetry on the live BOPTEST `bestest_air` runtime environment. The central question is not whether a controller can be trained on a surrogate, but which functional role each surrogate should play inside the learning loop.

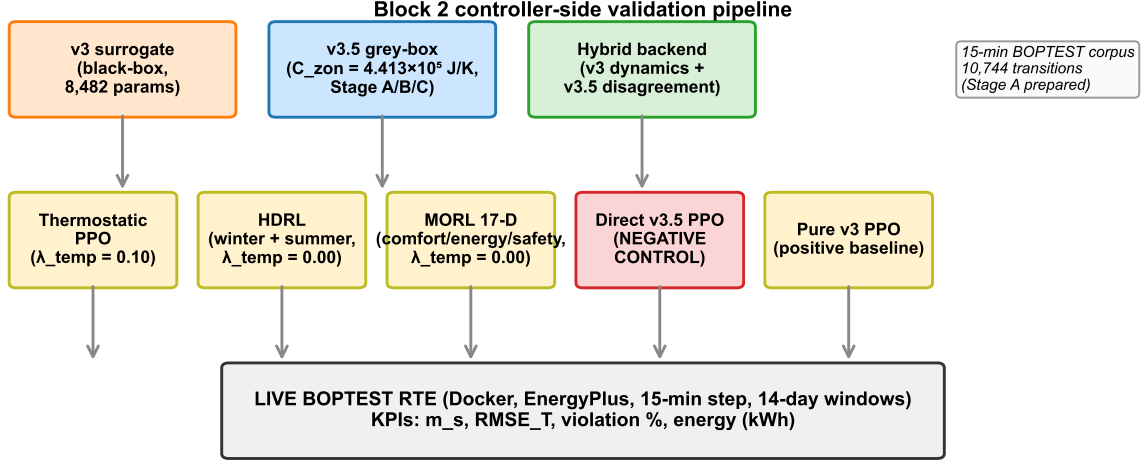
The experiments compare five controller families: pure-v3 thermostatic PPO, direct-v3.5 PPO, hybrid-v3/v3.5 PPO, hierarchical DRL (HDRL), and preference-conditioned MORL. All policies are trained on surrogate backends and then evaluated in closed loop against BOPTEST. Two targeted 14-day windows are used for the main thermostatic/HDRL comparison: `peak_heat_window` (January, daily-mean ambient -24.4°C) and `typical_heat_window` (February, $+2.4^{\circ}\text{C}$). MORL and PI reference values additionally use the 12-month yearly evaluation protocol. This difference is intentional: the targeted windows expose controller-family mechanisms, while the yearly protocol exposes seed stability and preference robustness.

Figure 1 summarizes the pipeline. Direct v3.5 is a negative control, not a failed implementation: it asks whether the highest-fidelity predictive surrogate can be used directly as the policy rollout environment (it cannot). The hybrid backend then asks whether the same physical model becomes useful when its role changes from dynamics provider to reward-shaping censor — yes for thermostatic PPO, no for HDRL at the same λ_T , and conditionally yes for MORL once the observation interface is widened from 5D to 17D.

Roadmap boundary and executed path. Block 2 evaluates which functional role each surrogate should play inside the learning loop, strictly in the order of `roadmap.md` Block 2:

1. train the pure-v3 thermostatic PPO baseline (§4);
2. run the direct-v3.5 warm-start negative control (§4.5);
3. run the thermostatic hybrid λ_T sweep (§5) and the three-step transfer-gap diagnostics (§5.5);
4. run the HDRL λ_T sweep (§6);
5. run the MORL 5D→17D observation ablation, Pareto sweep, and N=5 canonical seed analysis (§6.5–§9);
6. benchmark the PI reference and rebuild the Block 2 evidence tables (§10–§11).

Artifact provenance and rebuild commands for every table and figure are catalogued in roadmap Section 11.1.



All controller-side KPIs measured on the live BOPTEST RTE, not on the surrogate. Surrogate is used during training only.

Figure 1: Block 2 controller-learning pipeline. The experiments separate the rollout model, the physical regularizer, the controller family, the observation interface, and live BOPTEST validation.

Table 1: Nomenclature for Block 2.

Symbol	Unit	Meaning
m_s	–	BOPTEST-style maintenance score (lower is better; combines comfort violation and tracking)
$RMSE_T$	$^{\circ}\text{C}$	live closed-loop zone-temperature RMSE
Violation	%	fraction of steps outside the 21–24 C comfort band
λ_T, λ_P	–	hybrid temperature / power disagreement weights
Δm_s	–	live-minus-surrogate m_s transfer gap
g_a	–	L2 action-gap norm (surrogate vs live)
CV	–	coefficient of variation (std/mean) of m_s over seeds
w_c, w_e, w_s	–	MORL comfort / energy / safety preference weights

Table 2: Targeted-window and yearly scenario definitions (data: roadmap Section 11.1).

Scenario	Day idx	Start (s)	Dur. (d)	Ambient ($^{\circ}\text{C}$)	Role
peak_heat_window	3	259200	14	-24.4	January coldest, heating stress test
typical_heat_window	37	3196800	14	2.4	February moderate, deployment-realistic
yearly evaluation	12 months	—	14/mo	varied	MORL + PI yearly summary

Block 2 is structured as a sequence of falsifiable hypotheses, each tied to a roadmap section and resolved against live BOPTEST evidence (Table 3). This pre-registration-style framing is what lets the negative results (direct v3.5, HDRL over-regularization, MORL seed variance) be read as decisions rather than tuning accidents.

Table 3: Block 2 falsifiable-hypothesis ledger. Each hypothesis is tested on the live BOPTEST RTE in the indicated roadmap section.

	Hypothesis	roadmap	Verdict
H1	The higher-fidelity twin (v3.5) used directly as the rollout environment yields the best controller.	§4–4.5	FALSIFIED (direct v3.5 $m_s > 1.0$)
H2	Calibrated v3.5 is useful when its role changes from dynamics provider to a frozen reward-shaping censor.	§5	SUPPORTED (hybrid is best)
H3	The thermostatic-optimal censor weight $\lambda_T = 0.10$ transfers to the HDRL family.	§6	FALSIFIED (HDRL best at $\lambda_T = 0$)
H4	MORL viability is determined by the reward scalarization alone.	§6.5–7	FALSIFIED (needs the 17D observation)
H5	The single-seed MORL canonical ($m_s \approx 0.10$) is representative.	§8–9	FALSIFIED at N=5 (best of five; CV 0.42–0.61)

5.1 PPO interface, observation design, and reward

All Block 2 policies use Proximal Policy Optimization (PPO). The shared PPO settings are learning rate 3×10^{-4} during surrogate pretraining, discount factor $\gamma = 0.99$ (an effective horizon of ~ 25 h at the 900 s step), and ten optimization epochs per rollout. The families differ in rollout length, minibatch size, and total timestep budget because the training scripts were developed per family rather than as a single hyperparameter ablation.

Table 4: PPO training configuration used in Block 2 (verified from the project training scripts and configuration files).

Parameter	Thermostatic PPO	HDRL	MORL pretrain	MORL finetune
Algorithm	PPO, MlpPolicy	PPO, MlpPolicy	PPO, MlpPolicy	PPO load+continue
Learning rate	3.0×10^{-4}	3.0×10^{-4}	3.0×10^{-4}	1.0×10^{-4}
n_steps	1024	1024	2048	inherited
batch_size	4096	2048	64	inherited
n_epochs	10	10	10	10
γ	0.99	0.99	0.99	0.99
GAE λ	0.95	0.95	0.95	0.95
Clip range	0.20	0.20	0.20	0.20
Total timesteps	10M	12M	2M	100k
Seed	42	42	42	42–46 (N=5)

Thermostatic PPO baseline (roadmap §4). Block 2 opens with the control baseline: a single-level thermostatic PPO controller trained purely on the v3 rollout surrogate (the canonical 1-hour-step checkpoint `rc_node_v3_tsupply.pt`) with no v3.5 regularization ($\lambda_T = \lambda_P = 0$). The actor and critic are separate MLP heads (Stable-Baselines3 `MlpPolicy`): the actor maps the 17D observation (1) to a tanh-squashed continuous supply-temperature command (2), while the critic estimates the state value entering the PPO advantage. Training runs for 10M surrogate steps under the shared settings of Table 4; with entropy coefficient zero the evaluated policy is deterministic. Crucially, the trained policy is run *zero-shot* on the live BOPTEST RTE with no live finetuning, so the baseline is already a strict surrogate-to-simulator transfer test. It is the reference against which the two negative controls (direct v3.5, warm-start) and the hybrid are measured.

The canonical observation interface is the 17-dimensional extended TSup-style vector:

$$s_t = [x_t^{\text{phys}}, x_t^{\text{time}}, \hat{T}_{t+1:t+24h}^{\text{amb}}, a_{t-1}, \Delta T_t^{\text{zone}}] \in \mathbb{R}^{17}. \quad (1)$$

It contains 5 physical states (zone temperature, CO₂, clipped-log power, previous supply temperature, ambient), 4 cyclic time features, 5 ambient forecasts (+1, +3, +6, +12, +24 h), 2 previous-action terms,

and 1 causal-smoothed ΔT_{zone} . The failed MORL baseline uses only the earlier 5D observation, which lacks sufficient actuation and forecast context.

Table 5: Extended 17D observation feature groups (verified from `envs/tsup_features.py`).

Feature group	Dim.	Contents
Physical state (basic)	5	T_{zone} , CO_2 , clipped-log power, prev. T_{sup} , T_{amb}
Cyclic time	4	hour and day sine/cosine encodings
Ambient forecast	5	T_{amb} at +1, +3, +6, +12, +24 h
Previous action	2	$(a_{T_{\text{sup}}}, a_{\text{fan}})$ from last step
Temperature delta	1	causal-smoothed ΔT_{zone}
Total (extended)	17	obs_mode = extended

The action is a single normalized supply-temperature command,

$$a_t \in [-1, 1], \quad T_t^{\text{sup}} = 18 + \frac{a_t + 1}{2}(35 - 18) \quad [^\circ\text{C}], \quad (2)$$

with a 1.0 C deadband and a per-step rate limit; the comfort band is 21–24 °C.

Evaluation metric. The headline maintenance score combines the comfort-violation rate and the worst-case relative severity over a rollout,

$$m_s = r_{\text{time}} + r_{\text{sev}}, \quad r_{\text{time}} = \frac{1}{N} \sum_{t=1}^N \mathbf{1}[T_t < T_\ell \text{ or } T_t > T_h], \quad (3)$$

$$r_{\text{sev}} = \max_t \max \left(\frac{(T_\ell - T_t)_+}{T_\ell}, \frac{(T_t - T_h)_+}{T_h} \right).$$

with $T_\ell = 21^\circ\text{C}$ and $T_h = 24^\circ\text{C}$ (source: `evaluation/benchmark_bestest_air_article7_style.py`). Hence r_{time} is the fraction of steps outside the band (violation % = 100 r_{time}) and r_{sev} is the single worst relative band exceedance; lower m_s is better. RMSE_T is reported against the band center $T^* = 22.5^\circ\text{C}$.

Table 6: Reward-shaping parameters (verified from `configs/env.yaml`).

Component	Value	Role
Comfort band	21.0 / 24.0 C	comfort interval (band_low/high)
Comfort deadband	0.50 C	soft margin around band edges
In-band bonus	0.050/step	reward for staying inside the band
Undershoot / overshoot weight	1.15 / 1.15	generic violation penalties
Cold-ambient asymmetry	1.60 ($\text{amb} < 8.0^\circ\text{C}$)	extra cold-undershoot penalty
Hot-ambient asymmetry	1.80 ($\text{amb} > 24.0^\circ\text{C}$)	extra hot-overshoot penalty
Heating action bonus	0.04 ($T_{\text{sup}} \geq 29.0^\circ\text{C}$)	anti-degenerate shaping
Cooling action bonus	0.06 ($T_{\text{sup}} \leq 21.0^\circ\text{C}$)	anti-degenerate shaping
MORL weights ($w_c/w_e/w_s$)	0.80 / 0.20 / 0.00	canonical scalarization
Energy scale	$2e - 4$ (W→reward)	energy-to-reward conversion

The per-step comfort term is piecewise linear with an asymmetric, ambient-dependent slope (source `envs/backends/bopetest_backend.py`):

$$r_t^{\text{comfort}} = \begin{cases} -w_u (T_\ell - T_t) + b_{\text{heat}} & T_t < T_\ell, \\ +\beta & T_\ell + \delta \leq T_t \leq T_h - \delta, \\ -w_o (T_t - T_h) + b_{\text{cool}} & T_t > T_h, \end{cases} \quad (4)$$

where $w_u = 1.60$ when $T_{\text{amb}} \leq 8^\circ\text{C}$ (else 1.15), $w_o = 1.80$ when $T_{\text{amb}} \geq 24^\circ\text{C}$ (else 1.15), the in-band bonus is $\beta = 0.05$, the deadband is $\delta = 0.5^\circ\text{C}$, and $b_{\text{heat}}, b_{\text{cool}}$ are small action-direction bonuses. The energy term is $r_t^{\text{energy}} = -\eta P_t$ with $\eta = 2 \times 10^{-4}$, and MORL scalarizes the objectives as $r_t = w_c r_t^{\text{comfort}} + w_e r_t^{\text{energy}} + w_s r_t^{\text{safety}}$. The asymmetric cold/hot weights encode that recovering a cold zone in cold weather is physically harder than the symmetric reverse.

5.2 Hybrid backend: mathematical role of v3.5

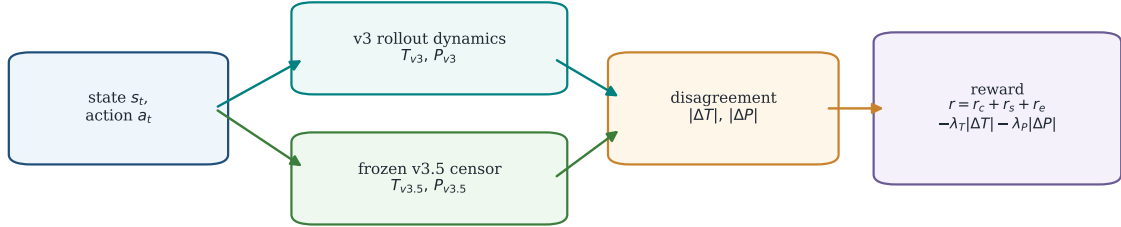
The hybrid backend changes the role of the calibrated physical surrogate. Instead of rolling out the policy on v3.5 directly, PPO rolls out on v3 and evaluates frozen v3.5 in parallel on the same state-action pair. The per-step reward is augmented by a disagreement penalty:

$$r_t^{\text{hyb}} = r_t^{\text{comfort}} + r_t^{\text{smooth}} + r_t^{\text{energy}} - \lambda_T |T_{t+1}^{v3} - T_{t+1}^{v3.5}| - \lambda_P |P_{t+1}^{v3} - P_{t+1}^{v3.5}|. \quad (5)$$

For the canonical thermostatic hybrid, $\lambda_T = 0.10$ and $\lambda_P = 5.0 \times 10^{-5}$; PPO otherwise computes the advantage $A_t = r_t^{\text{hyb}} + \gamma V(s_{t+1}) - V(s_t)$ unchanged. Thus v3.5 is neither a second policy loss nor a direct dynamics model: it is a frozen physics-informed censor that discourages the policy from entering state-action regions where the smooth v3 rollout and the calibrated physical model disagree. Across the canonical hybrid traces (`reports/hybrid_disagreement_summary.csv`, overall), the mean temperature disagreement is 0.969°C (p95 2.52°C) and the mean power disagreement is 708 W (p95 1236 W) — large enough to shape learning, bounded enough to stay informative.

Hybrid backend: per-step reward shaping

v3.5 is a frozen reward-shaping censor --- NOT a policy-loss term and NOT the rollout model



canonical thermostatic: $\lambda_T = 0.10$, $\lambda_P = 5 \times 10^{-5}$ | measured disagreement: mean $|\Delta T| = 0.969^\circ\text{C}$, mean $|\Delta P| = 708\text{ W}$

PPO rolls out only on v3; v3.5 is evaluated in parallel and enters the scalar reward, so the advantage A_t uses the shaped reward unchanged.

Figure 2: Hybrid reward-shaping mechanism. v3 supplies rollout dynamics; frozen v3.5 supplies a disagreement penalty on the same state-action transition.

5.3 Thermostatic PPO: direct-v3.5 failure and hybrid success (roadmap §4–§5)

Table 7 and Figure 3 summarize the main Block 2 controller result. Pure v3 PPO is already usable ($m_s = 0.073$ peak, 0.095 typical). Direct v3.5 PPO fails catastrophically despite v3.5’s superior Block 1 predictive fidelity: live violation reaches 77.1% on peak and 82.4% on typical, with RMSE above 4.3°C . The hybrid backend resolves the conflict: $m_s = 0.087$ on peak and 0.041 on typical, with violation below 5% on both windows and lower energy than pure v3 on the peak window.

Table 7: Canonical Block 2 live BOPTEST controller comparison on the targeted 14-day windows. Data sources for every row are catalogued in roadmap Section 11.1.

Policy/backend	Scenario	m_s	Violation (%)	RMSE _T ($^\circ\text{C}$)	Energy (kWh)
pure v3 PPO	peak	0.073	1.49	0.869	322.2
direct v3.5 PPO	peak	1.046	77.08	4.320	556.2
hybrid $\lambda_T = 0.10$	peak	0.087	4.69	0.795	305.3
pure v3 PPO	typical	0.095	4.39	0.622	368.3
direct v3.5 PPO	typical	1.102	82.37	4.401	580.5
hybrid $\lambda_T = 0.10$	typical	0.041	2.38	0.633	352.8

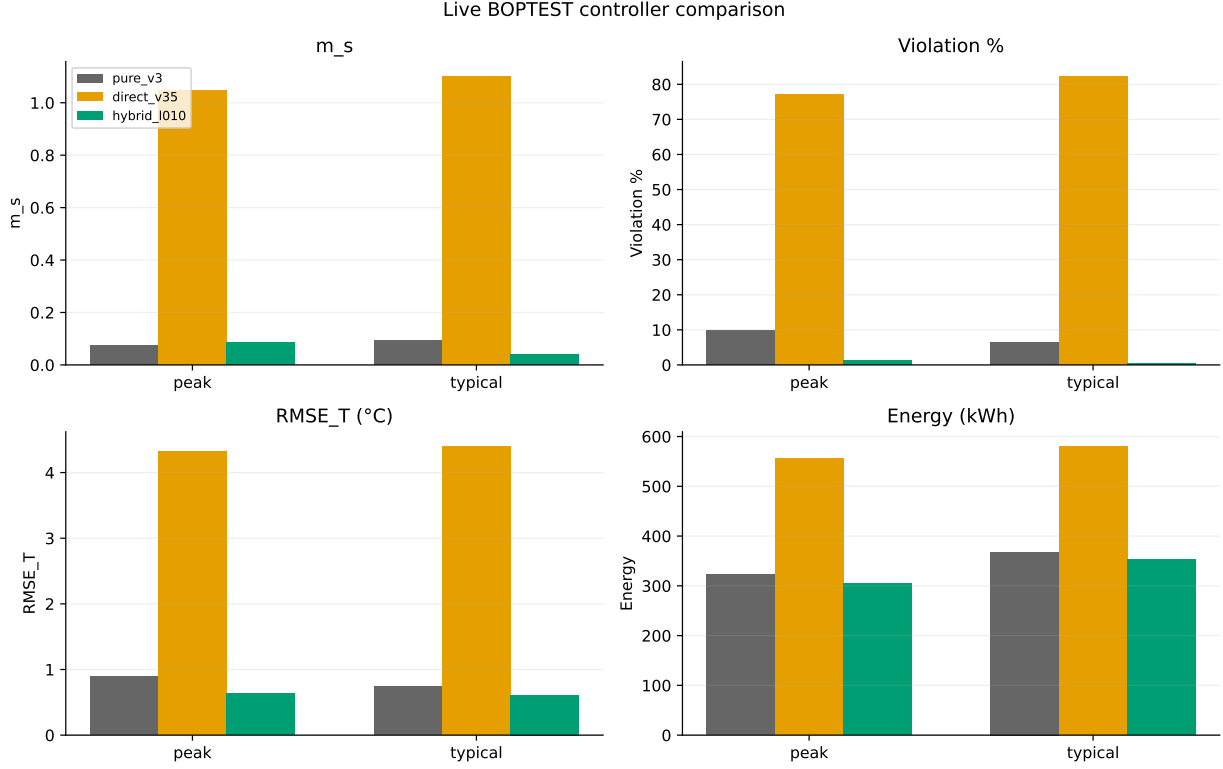


Figure 3: Live BOPTEST KPI comparison for pure v3, direct v3.5, and hybrid PPO. Direct v3.5 is a negative control: higher predictive fidelity does not imply a useful RL training environment.

Reading m_s step by step. The decomposition $m_s = r_{\text{time}} + r_{\text{sev}}$ (Eq. 3) exposes the failure structure directly (Figure 4). Direct v3.5’s peak $m_s = 1.046$ is dominated by $r_{\text{time}} \approx 0.77$ (out of band 77% of the time) plus a large worst-case severity $r_{\text{sev}} \approx 0.28$; the hybrid’s peak $m_s = 0.087$ is almost entirely a small r_{sev} with $r_{\text{time}} < 0.05$. The hybrid therefore does not merely lower the average error — it removes the sustained band departures that dominate the direct-v3.5 score.

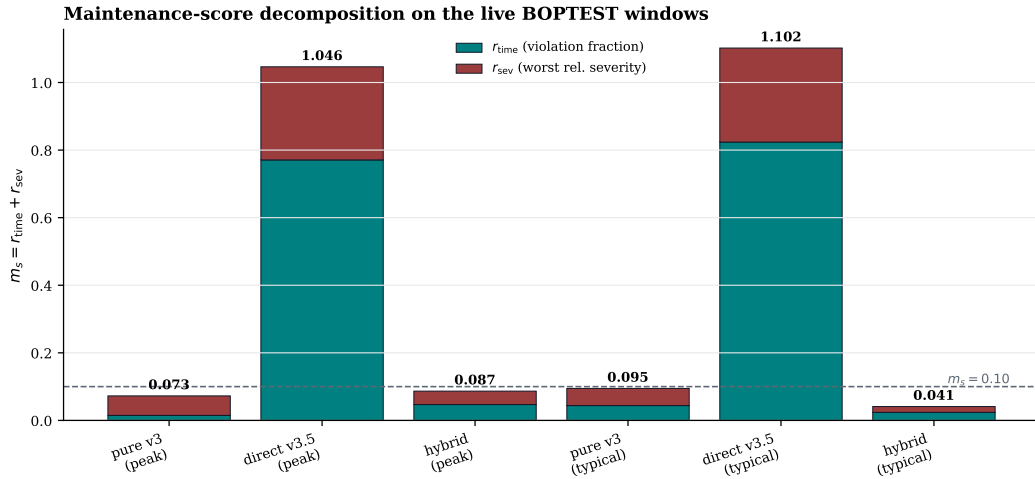


Figure 4: Data-driven decomposition $m_s = r_{\text{time}} + r_{\text{sev}}$ on the live BOPTEST windows. Direct v3.5 fails through sustained violation (r_{time}); pure v3 and the hybrid keep both terms small.

The time-series and action diagnostics in Figures 5 and 6 explain the mechanism. Direct v3.5 learns a bang-bang-like control law that drives the live simulator outside the comfort band; in phase space it places extreme actions in temperature-error regimes the live building does not support. We note explicitly that this destabilization mechanism is *hypothesized* (higher advantage-estimator variance under sharper surrogate predictions, and/or overfitting to sub-step physical structure unusable at the

15-min cadence) and is not directly measured here; discriminating the two is deferred to future work.

Block 2 closed-loop traces with physical bounds and ambient disturbance

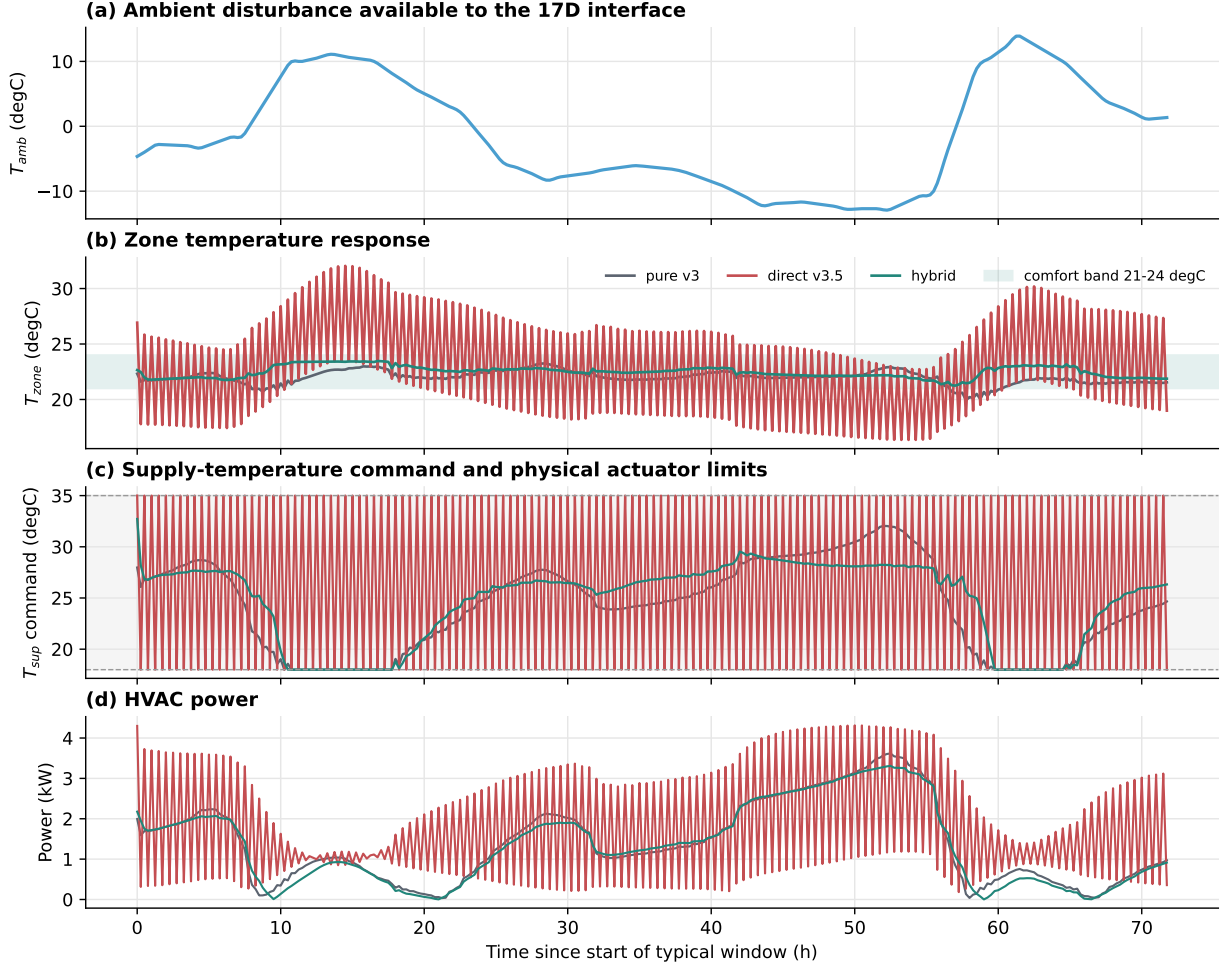


Figure 5: Closed-loop BOPTEST traces with ambient disturbance, comfort band, and physical actuator limits.

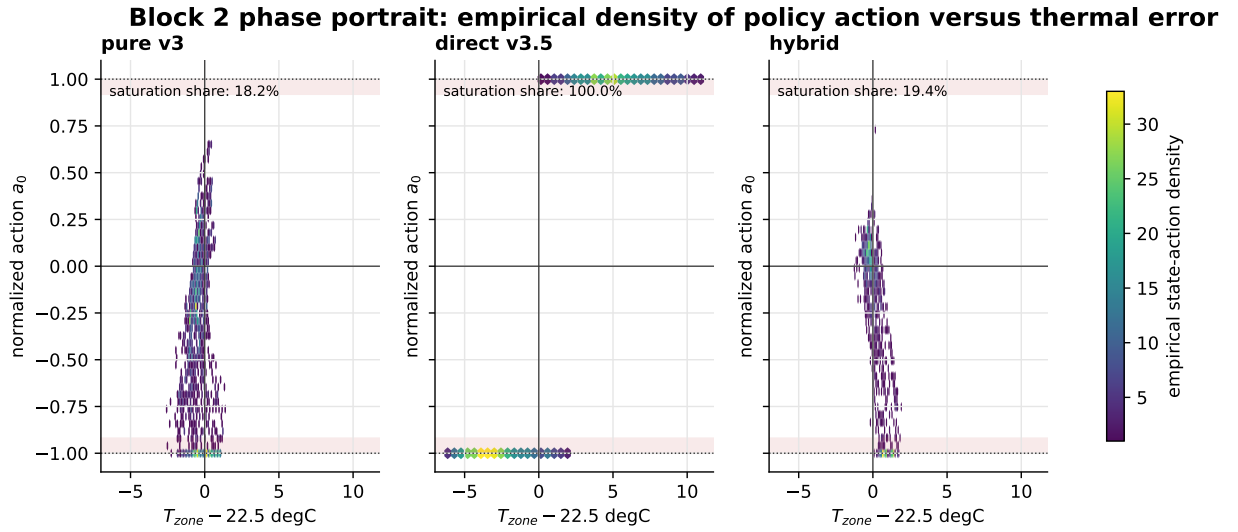


Figure 6: Phase portrait as empirical state-action density. The saturation bands near $a_0 = \pm 1$ expose the bang-bang behavior of direct v3.5 PPO.

Hybrid λ_T sweep and canonical selection (roadmap §5). The canonical hybrid operating point was selected by a thermostatic sweep over the temperature-disagreement weight $\lambda_T \in \{0.05, 0.10, 0.15\}$ at fixed $\lambda_P = 5 \times 10^{-5}$ (Table 8). The mid setting $\lambda_T = 0.10$ (**hybrid_1010**) is canonical: per the roadmap selection rule it retains the energy advantage over pure v3 while avoiding the stronger comfort degradation seen at the weaker (0.05) and stronger (0.15) censor settings. Only the canonical point’s live-BOPTEST KPIs are retained as a frozen artifact; the bracketing points served selection only.

Table 8: Thermostatic hybrid λ_T sweep design (roadmap §5; $\lambda_P = 5 \times 10^{-5}$ throughout). The mid point is the retained canonical.

λ_T	Tag	Role
0.05	hybrid_l005	weaker censor (sweep bracket)
0.10	hybrid_l010	canonical (retained; $m_s = 0.087$ peak, 0.041 typical)
0.15	hybrid_l015	stronger censor (sweep bracket)

5.4 Warm-start negative control (roadmap §4.5)

A second negative control tests whether v3.5 is useful as a policy *initializer* rather than a reward-shaping censor. The benchmark (`run_block2.py warmstart`) runs four steps: (1) *pretrain* a thermostatic policy on the calibrated v3.5 surrogate; (2) *scratch finetune* a thermostatic policy from random initialization directly on the live BOPTEST RTE; (3) *warm-start finetune* a second policy on BOPTEST initialized from the v3.5-pretrained checkpoint of step 1; and (4) *evaluate* both finetuned policies on the same BOPTEST windows and compare. The comparison is therefore internally controlled — scratch and warm-start differ only in initialization. It does not help: warm-started policies are markedly worse than the scratch finetune (Table 9), raising m_s by roughly two to three times on both windows. The problem is the role assigned to the surrogate during early policy formation, not a lack of fine-tuning; v3.5 belongs in the reward as a censor, not in the weights as an initializer.

Table 9: Direct-v3.5 warm-start utility (outputs/block2_thermostatic_warmstart_utility/comparison_summary.csv).

Mode	Scenario	m_s	Violation (%)
scratch (random init)	peak	0.465	31.70
scratch (random init)	typical	0.578	45.54
warm-start (from v3.5)	peak	1.270	83.93
warm-start (from v3.5)	typical	1.289	86.83

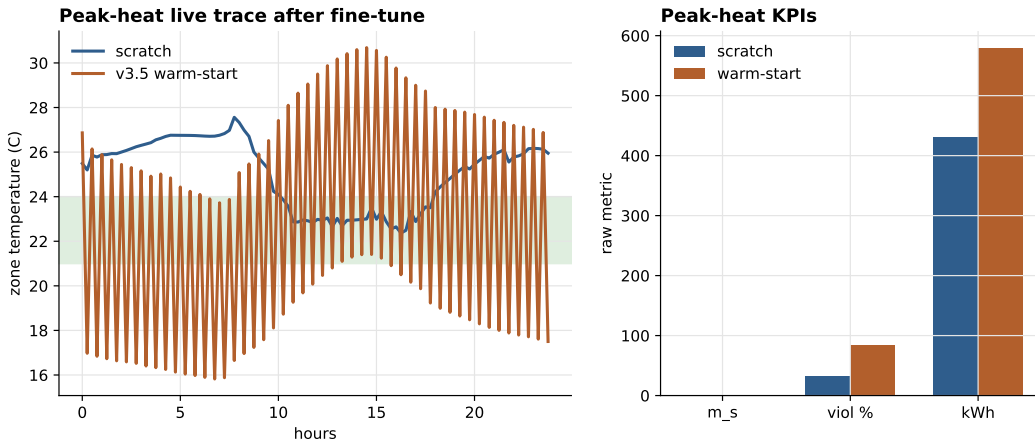


Figure 7: Warm-start negative control. Pretraining on direct v3.5 and then fine-tuning on the hybrid backend is inferior to training the hybrid policy from scratch.

5.5 Transfer-gap diagnostics (roadmap §5.5)

This diagnostic (roadmap §5.5) compares three controllers — pure v3, hybrid $\lambda_T = 0.10$, and direct v3.5 — in three steps. *Step 1*: a standalone direct-v3.5 thermostatic policy is trained on the calibrated v3.5 environment (the failure control; `thermostatic-train -variant v35_direct`), separate from the warm-start checkpoint of the previous section because it answers a different question (zero-shot live transfer rather than warm-start utility). *Step 2*: all three policies are run in closed loop on the live BOPTEST RTE for the two 14-day windows (`thermostatic-transfer -variant all`). *Step 3*: for each policy we compute four transfer diagnostics (`thermostatic-diagnose`) — the temperature tracking gap (live RMSE_T), the surrogate-vs-live maintenance-score mismatch

$$\Delta m_s = m_s^{\text{surrogate}} - m_s^{\text{BOPTEST}} \quad (\text{negative} \Rightarrow \text{surrogate optimistic}),$$

$$g_a = \frac{1}{N} \sum_{t=1}^N \|a_t^{\text{BOPTEST}} - a_t^{\text{surrogate}}\|_2. \quad (6)$$

the action-gap norm g_a , and the first divergence step (earliest 15-min step where surrogate and live trajectories separate beyond threshold), with the top driver feature attributed to the divergence.

Direct v3.5 has the largest transfer mismatch ($|\Delta m_s| \approx 0.9\text{--}1.0$, $g_a \approx 2.0$, live $\text{RMSE}_T > 4.3^\circ\text{C}$), and its top divergence driver is t_{zone} : its sharp temperature dynamics produce live actions unlike the surrogate rollout. Hybrid $\lambda_T = 0.10$ has the smallest gap ($|\Delta m_s| \approx 0.02$) and, on the typical window, holds the BOPTEST-consistent action for 16 steps before drifting (Table 10). The action-gap norm is the most diagnostic column: $g_a = 2.0$ for direct v3.5 means the policy sits at the opposite action bound from what BOPTEST would choose throughout the episode, i.e. a learned bang-bang exploit of the surrogate.

Table 10: Transfer diagnostics across three backends. $\Delta m_s = \text{surrogate} - \text{live}$ (negative $\Rightarrow$ surrogate optimistic); Temp gap is the live closed-loop RMSE_T (data: roadmap Section 11.1).

Variant	Scenario	Temp gap ($^\circ\text{C}$)	Δm_s	Action gap	First div.	Top driver
pure v3	peak	0.894	-0.102	0.377	1	p_total_norm
pure v3	typical	0.745	-0.068	0.333	1	p_total_norm
hybrid $\lambda_T = 0.10$	peak	0.633	-0.023	0.473	1	p_total_norm
hybrid $\lambda_T = 0.10$	typical	0.612	-0.021	0.253	16	p_total_norm
direct v3.5	peak	4.320	-0.886	2.000	1	t_zone_norm
direct v3.5	typical	4.401	-1.014	2.014	1	t_zone_norm

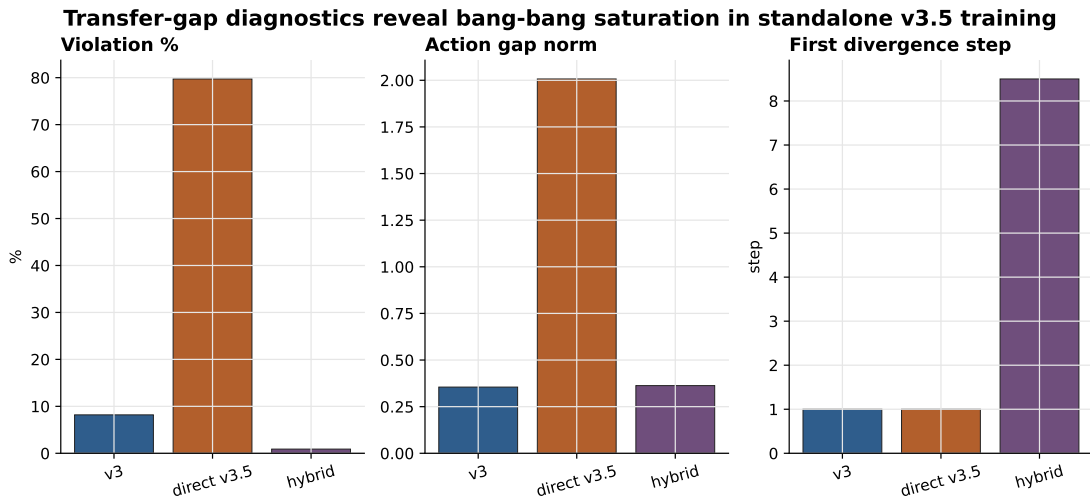


Figure 8: Transfer-gap diagnostics. Direct v3.5 has high action-gap norm and live-surrogate mismatch; the hybrid backend suppresses both.

5.6 HDRL sensitivity: the hybrid weight is controller-family specific (roadmap §6)

HDRL here is a *seasonal* hierarchy. A high-level seasonal selector routes control to one of two low-level PPO setpoint specialists — a winter agent (trained 5M steps on cold-season day ranges with cold-biased comfort shaping) and a summer agent (7M steps on warm-season ranges) — each acting on the same 17D observation and supply-temperature action. Formally the controller is $\pi(a_t | s_t) = \pi_{k(t)}(a_t | s_t)$ with the high-level gate $k(t) \in \{\text{winter}, \text{summer}\}$ chosen by season; the hierarchy supplies regime-aware temporal abstraction, so each low-level specialist solves a narrower, better-conditioned control problem than a single global policy (Figure 9).

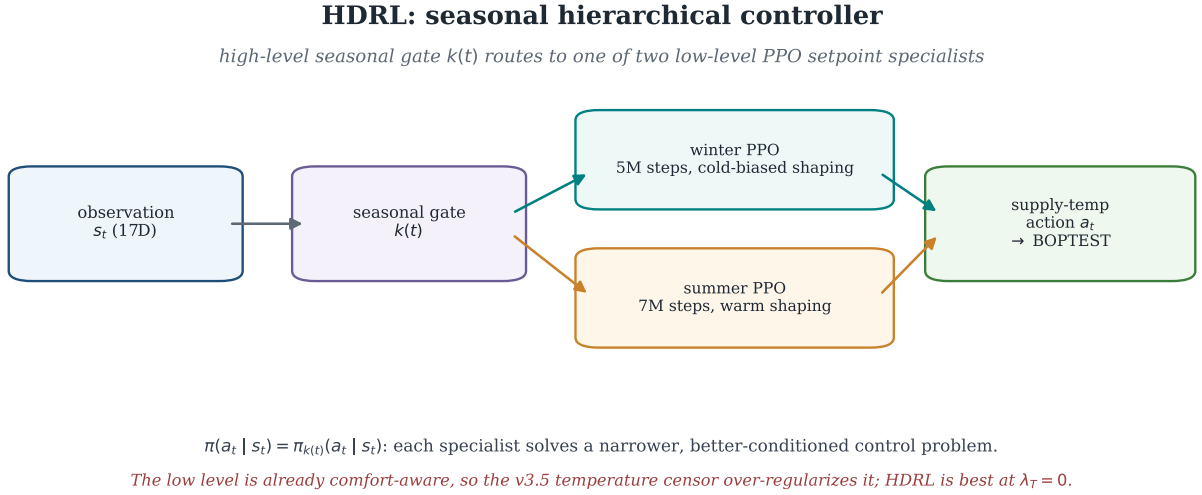


Figure 9: HDRL architecture: a high-level seasonal gate $k(t)$ routes the 17D observation to one of two low-level PPO setpoint specialists (winter / summer). Because the low level is already comfort-aware, the v3.5 temperature sensor over-regularizes it.

The HDRL experiment asks whether the thermostatic hybrid setting $\lambda_T = 0.10$ transfers to this hierarchical controller. It does not (Table 11). The mechanism is over-regularization: because each seasonal specialist already encodes comfort-aware structure (season-tuned shaping plus the high-level gate), the additional v3.5 temperature-disagreement censor constrains an already comfort-aware low-level loop, biasing it toward conservative under-heating; performance therefore degrades monotonically with λ_T , and the correct transfer keeps only the power channel ($\lambda_T = 0$). HDRL performs best at $\lambda_T = 0.00$ on both windows and degrades monotonically as temperature-disagreement regularization is increased. This shows the correct physical-censor strength depends on the controller family and its action decomposition, not on a universal weight.

Table 11: HDRL sweep over λ_T on the targeted windows; best is $\lambda_T = 0$, and performance degrades monotonically as the temperature censor strengthens (data: roadmap Section 11.1).

λ_T	Scenario	m_s	Violation (%)	RMSE _T (°C)	Energy (kWh)
0.00	peak	0.180	6.10	0.751	329.6
0.03	peak	0.307	11.90	0.993	311.5
0.05	peak	0.418	20.98	1.303	298.1
0.10	peak	0.440	22.99	1.343	300.4
0.00	typical	0.234	3.12	0.691	385.1
0.03	typical	0.296	9.38	0.959	369.6
0.05	typical	0.512	27.38	1.491	354.5
0.10	typical	0.511	30.65	1.455	357.1

Engineering reading. As λ_T rises, energy falls slightly (the censor biases the low-level loop toward conservative under-heating, e.g. $329.6 \rightarrow 300.4$ kWh on peak) but comfort collapses — peak violation roughly quadruples ($6.1\% \rightarrow 23.0\%$) and RMSE_T nearly doubles. HDRL already supplies comfort-aware temporal abstraction through its scheduler, so an additional temperature-disagreement censor is redundant and over-constrains the setpoint controller; the correct transfer of the Block 1 recipe to HDRL keeps only the power channel ($\lambda_T = 0$, $\lambda_P = 5 \times 10^{-5}$).

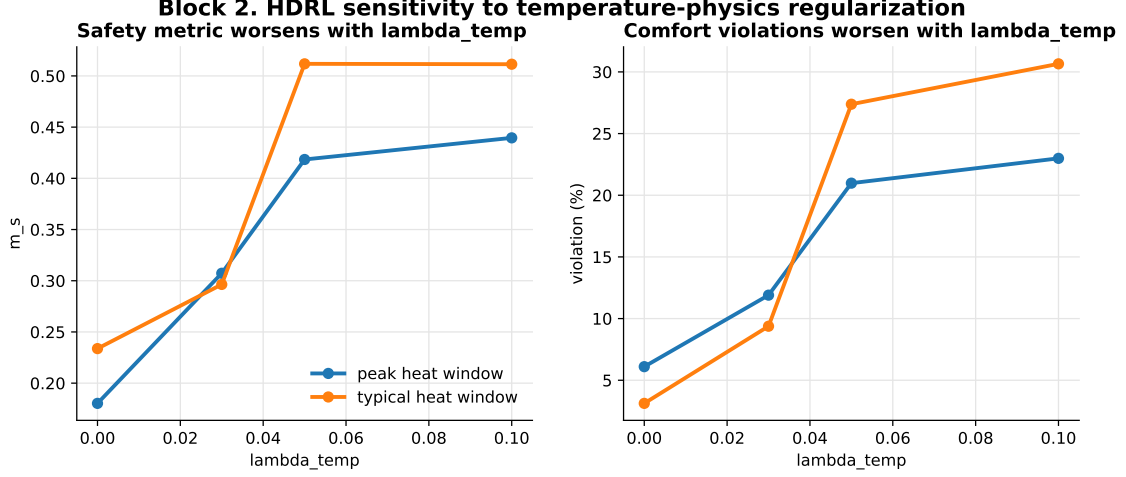


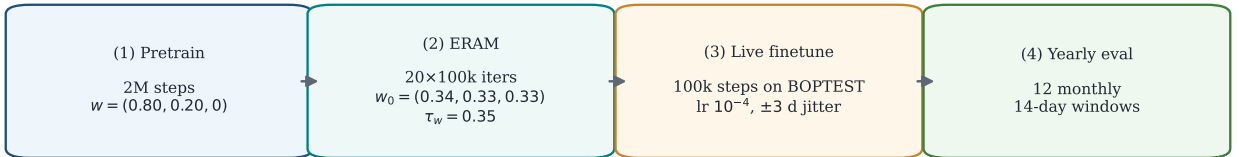
Figure 10: HDRL λ_T sensitivity. The thermostatic regularization weight does not transfer; the best HDRL policy uses no temperature-disagreement term.

5.7 MORL observation ablation: 5D failure to 17D success (roadmap §6.5–§7)

MORL uses a four-stage pipeline that differs from the single-stage PPO families (roadmap Section 7): (1) a 2M-step surrogate *pretrain* on the 17D hybrid backend with canonical $(w_c, w_e, w_s) = (0.80, 0.20, 0.00)$; (2) an *ERAM* weight-adaptation stage (20 iterations of 100k steps from initial weights $0.34/0.33/0.33$, weight-update temperature $\tau_w = 0.35$); (3) a 100k-step *finetune* on the live BOPTEST RTE at learning rate 10^{-4} with ± 3 -day episode-start jitter; and (4) a 12-month *yearly evaluation*. MORL is the only family with a live-BOPTEST finetune; the thermostatic/HDRL families are evaluated zero-shot after surrogate-only training, which is a strictly harder transfer (Figure 11).

MORL: preference-conditioned four-stage pipeline

17D observation + power-only hybrid backend ($\lambda_T = 0$, $\lambda_P = 5 \times 10^{-5}$)



preference weights $w = (w_c, w_e, w_s)$ condition the scalar reward $r = w_c r^c + w_e r^e + w_s r^s$;

MORL is the only family with a live-BOPTEST finetune; PPO/HDRL are evaluated zero-shot.

Figure 11: MORL four-stage pipeline: surrogate pretrain $\rightarrow$ ERAM preference adaptation $\rightarrow$ live-BOPTEST finetune $\rightarrow$ yearly evaluation, all on the 17D power-only hybrid backend. The preference vector conditions the scalarized reward.

MORL initially failed with a 5D observation (zone temperature, ambient, hour, day, occupancy). Under the current code path, a reconstructed 5D rerun obtains $\text{RMSE}_T = 2.721^\circ\text{C}$, violation 37.8%,

and $m_s = 0.680$; the originally frozen 5D artifact was even worse ($\text{RMSE}_T = 4.96^\circ\text{C}$, $m_s = 1.046$) and is retained only as an audit artifact. Replacing the observation with the 17D TSup-style vector recovers a usable policy: $\text{RMSE}_T = 0.720^\circ\text{C}$, violation 4.9%, $m_s = 0.099$ (Table 12). The dominant MORL bottleneck was the observation geometry, not the reward scalarization alone.

Table 12: MORL observation-interface ablation. The backend is hybrid in both cases; only the observation interface changes. The current-code reconstructed 5D rerun is the reproducible main-paper evidence (roadmap Section 11.1).

Variant	Obs dim	RMSE_T ($^\circ\text{C}$)	Violation (%)	m_s
MORL 5D (current-code rerun)	5	2.721	37.8	0.680
MORL 17D power-only (canonical)	17	0.720	4.9	0.099

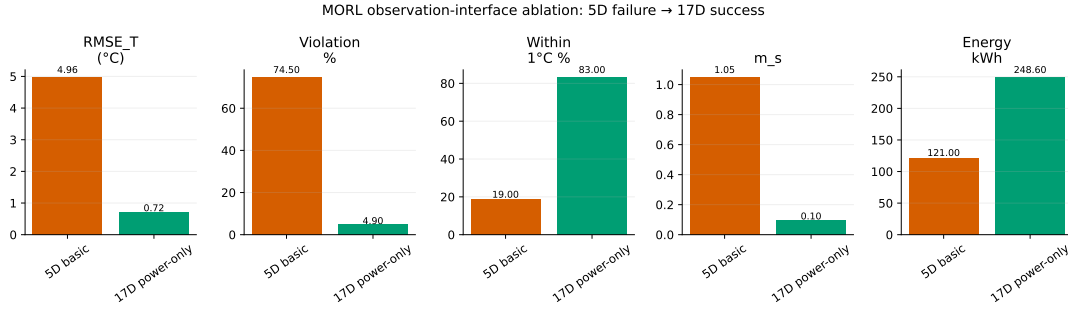


Figure 12: MORL 5D failure and 17D recovery. The observation interface, not only the scalarized reward, determines whether MORL is viable.

Power-only backend and claim discipline (roadmap §7, §9, §13). MORL inherits the controller-family lesson of the HDRL sweep: it runs on the *power-only* hybrid backend ($\lambda_T = 0$, $\lambda_P = 5 \times 10^{-5}$), because temperature-disagreement censoring hurts non-thermostatic families. The MORL claim is deliberately narrow and audit-protected. Before the seed expansion, commit 93df9b3 froze the canonical plan and the seed-45/46 falsification predictions; after $N = 5$, commit 62dc859 appended the observed variance without rewriting the pre-registration, and a replay test confirmed bit-identical BOPTEST trajectories for a fixed checkpoint (so the spread is training stochasticity, not simulator noise). The defensible reading is therefore: under the fixed 17D power-only backend, MORL is substantially stronger than the reconstructed 5D interface and yields a usable comfort–energy Pareto structure, but the $N = 5$ canonical variance (CV 0.42–0.61) is too high for a deployment-stability claim — the single-seed canonical is the best of five, not the median.

5.8 MORL Pareto front and N=5 seed variance (roadmap §8–§9)

The MORL Pareto sweep varies comfort/energy weights with safety weight zero (Table 13). Energy-only control collapses comfort ($m_s = 1.588$, violation 86.8%); comfort-leaning settings are far more stable. The single-seed (seed 42) Pareto points are reported separately from the $N = 5$ canonical extensions, because the seed analysis is the central audit result: the neutral 50/50 canonical has $m_s = 0.187 \pm 0.078$ (CV 0.42, 95% t -CI [0.090, 0.284]) and the practical 75/25 canonical has $m_s = 0.139 \pm 0.085$ (CV 0.61, 95% t -CI [0.033, 0.245]). Seed 46 is an outlier in both groups (Table 14). Because the replay audit produced bit-identical BOPTEST trajectories for a fixed policy, this variance is attributed to PPO/ERAM training stochasticity, not simulator noise. The single-seed canonical ($m_s \approx 0.10$) is therefore the *best* of five, not the median.

Table 13: MORL Pareto (seed 42) and N=5 canonical seed analysis (data: roadmap Section 11.1).

Preference / statistic	RMSE _T (°C)	Violation (%)	m_s	Interpretation
0/100 (seed 42)	8.359	86.76	1.588	energy-only collapse
25/75 (seed 42)	0.912	11.39	0.173	energy-weighted usable
50/50 (N=5 mean±std)	0.893±0.081	13.01±6.62	0.187±0.078	neutral, CV=0.42
75/25 (N=5 mean±std)	0.799±0.100	9.23±6.67	0.139±0.085	practical, CV=0.61
100/0 (seed 42)	0.631	1.49	0.032	best seed-42 comfort

Engineering reading. The front is strongly asymmetric. Energy-only control (0/100) collapses to a degenerate non-heating policy (86.8% violation, $m_s = 1.588$): minimizing energy simply means not heating. Comfort-only (100/0) is the best single-seed m_s but the most energy-hungry, and the practical 75/25 point recovers near-comfort-only comfort at lower energy — the recommended deployment compromise. The crucial caveat is the seed dimension: the single-seed front overstates reliability, because the $N = 5$ extension turns both canonicals into wide distributions (CV 0.42–0.61), so the Pareto curve should be read as a best-seed envelope, not a deployment guarantee.

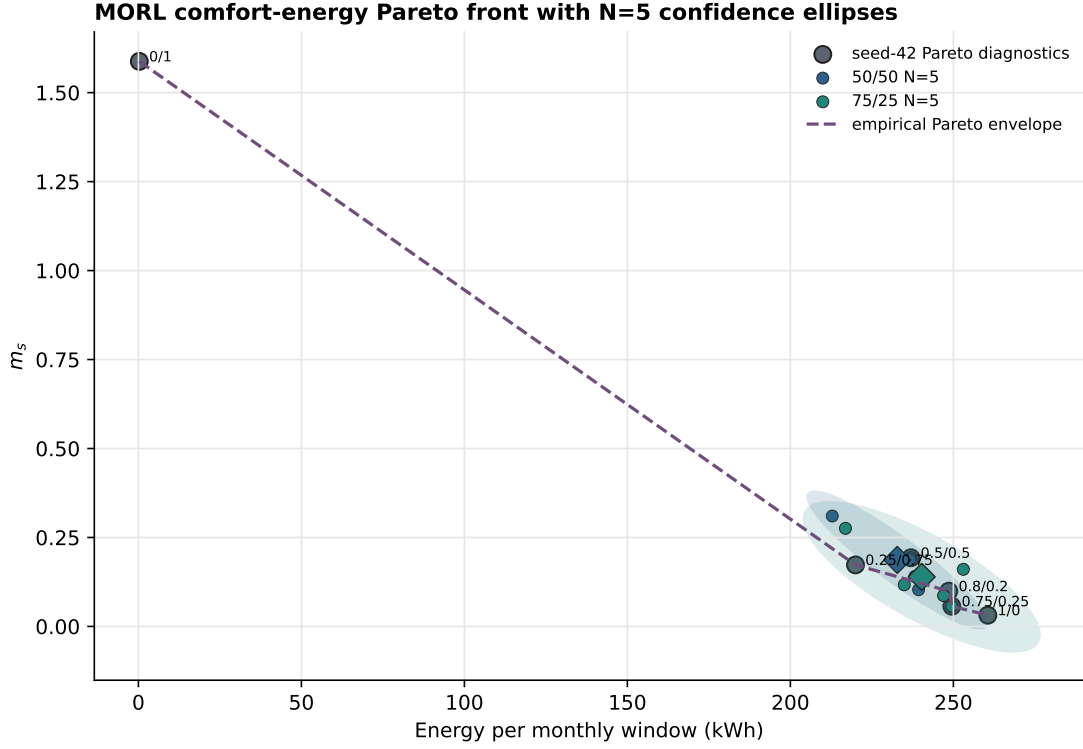
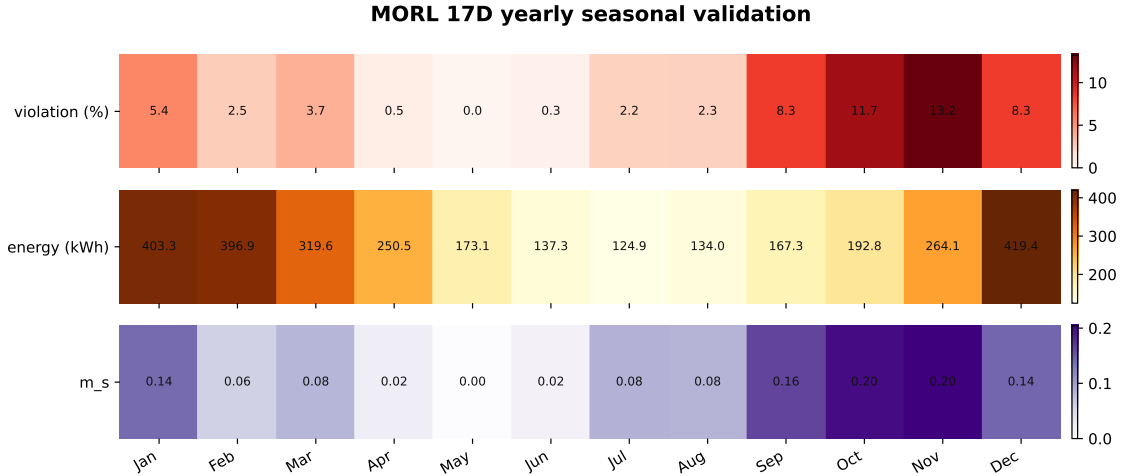
**Figure 13:** MORL comfort–energy Pareto front with N=5 confidence ellipses for the two canonical points. Non-canonical points are seed-42 only.

Table 14: Per-seed MORL yearly metrics for the two canonical weight pairs (data: roadmap Section 11.1).

Pair (c/e)	Seed	RMSE _T	Within 1°C (%)	Violation (%)	Energy (kWh)	m_s
50/50	42	0.939	74.9	12.92	2843.6	0.193
50/50	43	0.769	82.9	6.77	2872.0	0.103
50/50	44	0.879	76.9	9.34	2858.4	0.141
50/50	45	0.895	77.7	11.97	2839.8	0.187
50/50	46	0.985	65.4	24.05	2554.1	0.310
75/25	42	0.700	83.3	3.08	2994.3	0.057
75/25	43	0.716	84.9	5.47	2963.9	0.087
75/25	44	0.769	81.7	7.82	2819.1	0.117
75/25	45	0.903	78.5	9.42	3036.7	0.160
75/25	46	0.909	70.3	20.37	2602.8	0.276

5.9 Seasonal variance falsification (roadmap §9; audit §13)

At $N = 3$, the practical canonical appeared to show near-deterministic winter behavior and a seasonal inversion relative to the neutral canonical, motivating a pre-registered falsification test before seeds 45 and 46 were trained. The direction-specific predictions (February winter $\sigma(m_s) < 0.005$; June summer $\sigma(m_s) > 0.05$; winter neutral/practical variance ratio > 20) failed at $N = 5$: February practical $\sigma(m_s)$ rose to order 0.17 and the winter variance ratio collapsed to order one. This is a success of the pre-registration protocol (audit anchors 93df9b3 pre-registration, 62dc859 post- $N=5$ falsification), not a project failure. The honest, narrower conclusion: MORL is promising in mean performance, especially at comfort-leaning preferences, but is *not* deployment-stable without explicit stabilization (validation-based checkpoint selection, early stopping, or ensemble selection), which is left as future work because the canonical protocol fixes final-epoch evaluation.

**Figure 14:** MORL seasonal performance heatmap under the 17D interface. The figure shows monthly structure but does not support the $N=3$ mechanism claim after $N=5$ extension.

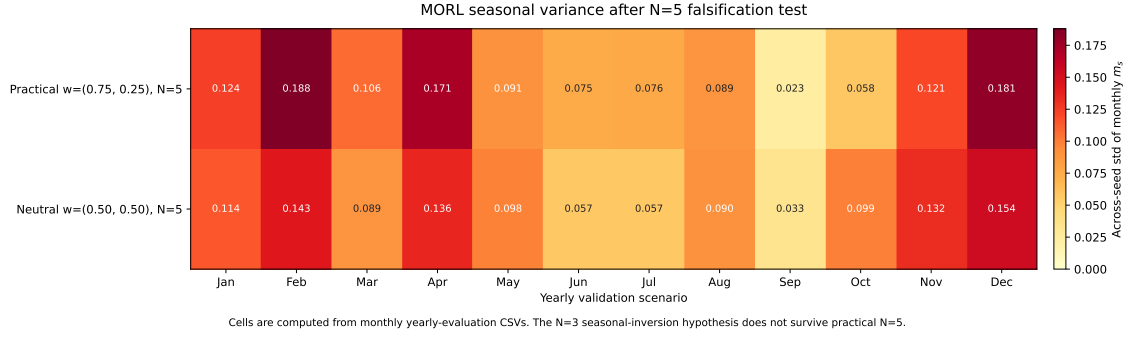


Figure 15: Post-N=5 seasonal variance diagnostic. The earlier seasonal-inversion hypothesis is reported as falsified, not retained as a positive mechanism claim.

5.10 PI reference and synthesis (roadmap §10–§11)

The BOPTEST built-in PI controller is a reproducible reference, not a tuned strong baseline. On the 12-month yearly evaluation it is comfort-poor but energy-frugal: mean $\text{RMSE}_T = 3.40^\circ\text{C}$, mean violation 63.6%, mean monthly energy 104.1 kWh, mean $m_s = 0.910$. All Block 2 RL/MORL agents reach m_s well below PI’s 0.910, dominating it on comfort at comparable or lower energy after accounting for the comfort/energy trade-off.

Table 15 consolidates the maintenance-score decomposition across every controller family. It makes the cross-family pattern explicit: usable controllers (pure v3, hybrid, MORL 17D) keep both r_{time} and r_{sev} small, whereas the failures (direct v3.5, PI) carry a large r_{time} — they spend a large fraction of the horizon outside the band. PI is the mirror image of direct v3.5: both have r_{time} near or above 0.6, but PI fails by chronic under-heating while direct v3.5 fails by saturated overshoot.

Table 15: Cross-controller maintenance-score decomposition $m_s = r_{\text{time}} + r_{\text{sev}}$ (Eq. 3). r_{time} = fraction of steps outside the band; r_{sev} = worst relative band exceedance. Thermostatic/HDRL rows are 14-day windows; MORL/PI rows are 12-month yearly means.

Controller (window/eval)	m_s	r_{time}	r_{sev}
pure v3 (peak)	0.073	0.015	0.058
direct v3.5 (peak)	1.046	0.771	0.276
hybrid $\lambda_T=0.10$ (peak)	0.087	0.047	0.040
HDRL $\lambda_T=0$ (peak)	0.180	0.061	0.119
pure v3 (typical)	0.095	0.044	0.051
direct v3.5 (typical)	1.102	0.824	0.278
hybrid $\lambda_T=0.10$ (typical)	0.041	0.024	0.017
HDRL $\lambda_T=0$ (typical)	0.234	0.031	0.202
MORL 50/50 (yearly, N=5)	0.187	0.130	0.057
MORL 75/25 (yearly, N=5)	0.139	0.092	0.047
PI (yearly)	0.910	0.636	0.274

Block 2 establishes four controller-side claims. First, predictive fidelity and RL training utility are not equivalent: direct v3.5 is the more accurate twin yet fails as a rollout environment (live violation $> 77\%$). Second, role separation works — v3 provides smooth rollout dynamics while frozen v3.5 acts as a physical censor through disagreement shaping — giving the canonical hybrid ($m_s = 0.087$ peak, 0.041 typical). Third, the censor strength is controller-family specific: HDRL rejects $\lambda_T = 0.10$ and is best at $\lambda_T = 0$. Fourth, MORL is viable only with the 17D interface, and its N=5 analysis reveals high seed variance and falsifies the N=3 seasonal-inversion mechanism. The engineering implication is that hybrid surrogate RL is a role-allocation problem — rollout smoothness, physical censoring, observation geometry, controller family, and seed stabilization are separate design axes. This is the bridge to Block 3, where the fixed `bestest_air` recipe is transferred to the hydronic BOPTEST family.

5.11 Limitations

- **Mechanism not measured.** The direct-v3.5 destabilization is established only in *direction*; the gradient-variance vs sub-step-overfit hypotheses are not discriminated here.
- **Single testcase.** All Block 2 results are on `bestest_air`; cross-building transfer is Block 3.
- **MORL seed stability.** The canonical MORL claim is narrowed: $N=5$ CV is 0.42–0.61; the single-seed canonical is the best of five and final-epoch evaluation is fixed by protocol.
- **Per-family PPO hyperparameters differ** (rollout length / batch size / budget); they were set per training script, so cross-family KPI differences are not a controlled hyperparameter ablation.
- **MORL alone uses live BOPTEST finetuning** (100k steps); thermostatic/HDRL are evaluated zero-shot after surrogate-only training, a strictly harder transfer.

Artifact provenance (which `reports/` and `outputs/` files back every table and figure) and the rebuild commands are documented in `roadmap.md` Section 11.1; this section is regenerated by `build_results2_overleaf.py`.

5.12 Results II conclusion

Block 2 converts the Block 1 surrogate asymmetry into a controller-design principle. The five hypotheses of Table 3 resolve into one coherent recipe: *v3 is always the rollout dynamics, v3.5 is always a frozen physical censor, and what changes across controller families is which channels of the v3.5 prediction are censored*. Direct use of the higher-fidelity twin as the rollout environment fails (live violation $> 77\%$, $g_a \approx 2.0$); using it as a reward-shaping censor succeeds for thermostatic PPO at $\lambda_T = 0.10$ ($m_s = 0.087/0.041$); the same temperature censor over-regularizes the comfort-aware HDRL hierarchy and must be switched off ($\lambda_T = 0$); and MORL is recovered only by widening the observation interface to 17D, after which a power-only censor and a comfort-leaning preference give the best mean behaviour — but with an audit-protected, deliberately narrowed claim because the $N = 5$ canonical variance (CV 0.42–0.61) is too high for a deployment-stability statement.

The maintenance-score decomposition (Table 15) gives the unifying physical reading: usable controllers keep both the violation-time term r_{time} and the worst-case severity r_{sev} small, whereas the two failure modes are mirror images — direct v3.5 fails by saturated overshoot and PI by chronic under-heating, both with r_{time} near or above 0.6. Engineering-wise, hybrid surrogate RL is therefore not a single architecture but a *role-allocation* problem across five separable axes: rollout smoothness, physical censoring, observation geometry, controller family, and seed stabilization. This recipe — the frozen `hybrid_1010` thermostatic policy and the MORL canonical — is the artifact carried into Block 3 (Results III), where it is transferred, with documented adapters, from `bestest_air` to the hydronic BOPTEST family.